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Bulk photovoltaic effect in a two-dimensional ferroelectric semiconductor α - In_2Se_3 †

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The bulk photovoltaic effect, arising from the separation of charge carriers driven by crystal symmetry, is an intriguing physical phenomenon that has been attracting broad interest in the field of photovoltaic applications due to its junction-free nature and potential to surpass the Shockley–Queisser limit. The photovoltaic applications of conventional ferroelectric materials with wide bandgaps (2.7–4 eV) are limited due to their low photocurrent densities and weak photovoltaic response in the visible light region. The emergence of two-dimensional ferroelectric semiconductors with coupled visible light absorption and spontaneous polarization characteristics is promising for the development of functional photoferroelectrics. Herein, we report the experimental demonstration of enhanced bulk photovoltaic response in a two-dimensional ferroelectric semiconductor α - In_2Se_3 under the excitation of visible light. The generated photovoltaic current density is nearly two orders of magnitude higher than that of conventional bulk ferroelectric materials. Our findings highlight the potential of two-dimensional ferroelectric semiconductor materials for bulk photovoltaic applications across a broad spectral region.

1. Introduction

The pursuit of advanced photovoltaic devices with a large open-circuit voltage, high power conversion efficiency, and low-cost fabrication has sparked intense research not only on novel materials for energy harvesting but also on new photoelectric conversion mechanisms.¹ In contrast to conventional semiconductor photovoltaic devices, which rely on PN junctions or external electrical fields for the separation of photo-

excited electron–hole pairs,² it has been discovered that bulk ferroelectric materials with non-centrosymmetric structures can achieve such a separation through intrinsic polarization electric fields, known as the bulk photovoltaic (BPV) effect.^{3–7} The BPV effect can be implemented in straightforward device structures with exceptional properties, such as generation of open-circuit voltage exceeding the bandgap⁸ and power conversion efficiency beyond the Shockley–Queisser limit.⁹ However, the utilization of BPV in conventional ferroelectric materials (BaTiO_3 , LiNbO_3 , etc.) has been restricted due to their bandgap in the ultraviolet (UV) region and low conductivity in bulk materials.^{10–12} In order to improve the BPV performance, various strategies have been proposed, such as bandgap reduction,^{13,14} thickness reduction of the ferroelectric layer,^{15,16} and optimization of ferroelectric-electrode interfaces.¹⁷

The recent advancements in two-dimensional (2D) van der Waals (vdWs) materials have sparked growing interest in the BPV effect. These materials encompass a diverse range of types (such as metals, semi-metals, semiconductors and insulators) and allow for exotic physical phenomena at the atomic-layer level, e.g., the nonlinear optical effect,¹⁸ ferroelectricity,^{19–21} and flexo-photovoltaic effect.^{22,23} The BPV response in 2D vdWs materials is expected to offer appealing characteristics, including a high photocurrent density, a wide BPV response range, and low energy consumption. In many 2D materials, such as MX ($M = \text{Ge}, \text{Sn}; X = \text{S}, \text{Se}$),²⁴ WTe_2 ,²⁵ In_2Se_3 ,^{26–28} CuInP_2S_6 ,²⁹ and 2D sliding ferroelectrics,³⁰ the BPV effect has been examined through theoretical predictions or experimental studies. Among them, α - In_2Se_3 , a 2D ferroelectric semiconductor, has attracted significant attention in the field of photoferroelectrics³¹ due to its thickness-dependent bandgap in the visible light region (1.36–2.5 eV),^{32,33} and intercorrelated in-plane and out-of-plane electrical polarizations.^{34–36} The monolayer α - In_2Se_3 has been theoretically predicted to have a shift current response of $750 \mu\text{A V}^{-2}$, higher than that of other 2D materials, and its response spectrum extends even into the near-infrared region.³¹ Moreover, α - In_2Se_3 exhibits a remarkable Curie temperature up to 500 K,^{31,36} enabling its bulk

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photovoltaic response to function effectively in high-temperature environments.

In this article, we present an experimental demonstration of the BPV effect in a thin graphite/ α -In₂Se₃/thin graphite device. The device exhibits an enhanced BPV photocurrent when illuminated with different laser wavelengths (457 nm, 514 nm, and 633 nm) within the visible light range, arising from its low-dimensional nature, strong out-of-plane polarization intensity, and narrow bandgap. Our findings indicate that 2D ferroelectric semiconductors hold great potential for the development of optoelectronic devices across a wide spectral region.

2. Results and discussion

In this work, we focus on hexagonal (2H) α -In₂Se₃ flakes (Fig. 1(a)), which possess a non-centrosymmetric structure³⁷ and exhibit relatively higher stability compared to the rhombohedral (3R) structure (Fig. S1†).³⁸ The Raman spectroscopy measurement indicates a 2H stacking configuration with a splitting peak at around 89 cm⁻¹ (Fig. S2†).³⁹ Each layer in a unit cell of α -In₂Se₃ bulk has five subatomic layers interconnected by covalent bonds in the order Se–In–Se–In–Se. The central Se atoms deviate from the centrosymmetric position in the covalent bond configuration, leading to both in-plane and out-of-plane polarizations (P'_{ip} and P'_{oop}). Then, an alternating retention of net in-plane polarization (P_{ip}) occurs with the increase in the layer number, resulting from the opposite directions of P'_{ip} in adjacent layers. In contrast, the net accumulation of out-of-plane polarization (P_{oop}) increases with the increase of the layer number, since P'_{oop} in each layer maintains the same direction. Relevant experiments also evidenced the presence of layer-dependent in-plane ferroelectricity in 2H α -In₂Se₃.⁴⁰ Thus, the electrical polarization of 2H α -In₂Se₃ flakes is primarily contributed by P_{oop} . In order to characterize

structural symmetry, we conducted measurements of second harmonic generation (SHG) using an excitation laser at both oblique and normal incidence. As shown in Fig. 1(b), a SHG signal can only be detected at oblique incidence when the light field partially interacts with P_{oop} . In addition, electrical transport measurements were conducted on a device consisting of thin graphite/ α -In₂Se₃/thin graphite. The current–voltage $|I_d|$ – V_d curve of the device exhibits a hysteretic behavior, indicating the presence of ferroelectricity of 2H α -In₂Se₃ (Fig. S3†).

In order to investigate the BPV response by measuring efficient carrier collection, a vertical heterostructure of thin graphite/2H α -In₂Se₃/thin graphite was fabricated analogous to conventional ferroelectric photovoltaic devices. As illustrated in Fig. 2(a), spontaneous polarization P (black arrow) along the z -axis in α -In₂Se₃ leads to the accumulation of polarization (bound) charges at both the top and bottom interfaces. Even though free charges from the graphite electrodes could compensate for partial polarization charges, residual polarization charges will induce a depolarization field E_{dp} , which in turn leads to the separation of photo-excited charge carriers. As shown in its energy band diagram, polarization charges at the interfaces alter the chemical potential of top and bottom graphite through doping with an equal number of free charges, leading to distinct vacuum levels at each interface of α -In₂Se₃. Therefore, photo-excited electrons and holes can be drifted to the top and bottom interfaces under the influence of E_{dp} and ultimately collected by the two graphite electrodes. In addition, a symmetric contact between the top and bottom electrodes in the device can effectively exclude short-circuit current from the Schottky barrier effect.

Fig. 2(b) displays the current–voltage (I – V) characteristics of the D1 device at room temperature under laser excitation (red curve) and dark (black curve) conditions. Short-circuit current I_{sc} and open-circuit voltage V_{oc} in I – V curves are obtained at zero voltage and zero current, respectively. A typical I_{sc} value of

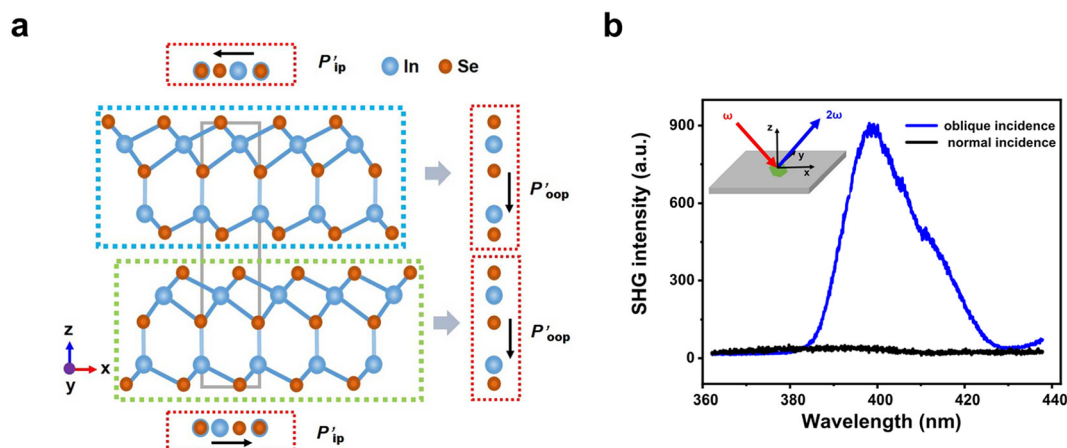


Fig. 1 Spontaneous polarization for 2H α -In₂Se₃. (a) Side view of bilayer hexagonal α -In₂Se₃ with In and Se atoms represented by blue and orange balls, respectively. The black arrows depict possible spontaneous polarizations. (b) SHG signal generated from an In₂Se₃ flake at normal incidence (blue curve) and oblique incidence (black curve), respectively. The inset shows the schematic diagram of light illumination and SHG generation.

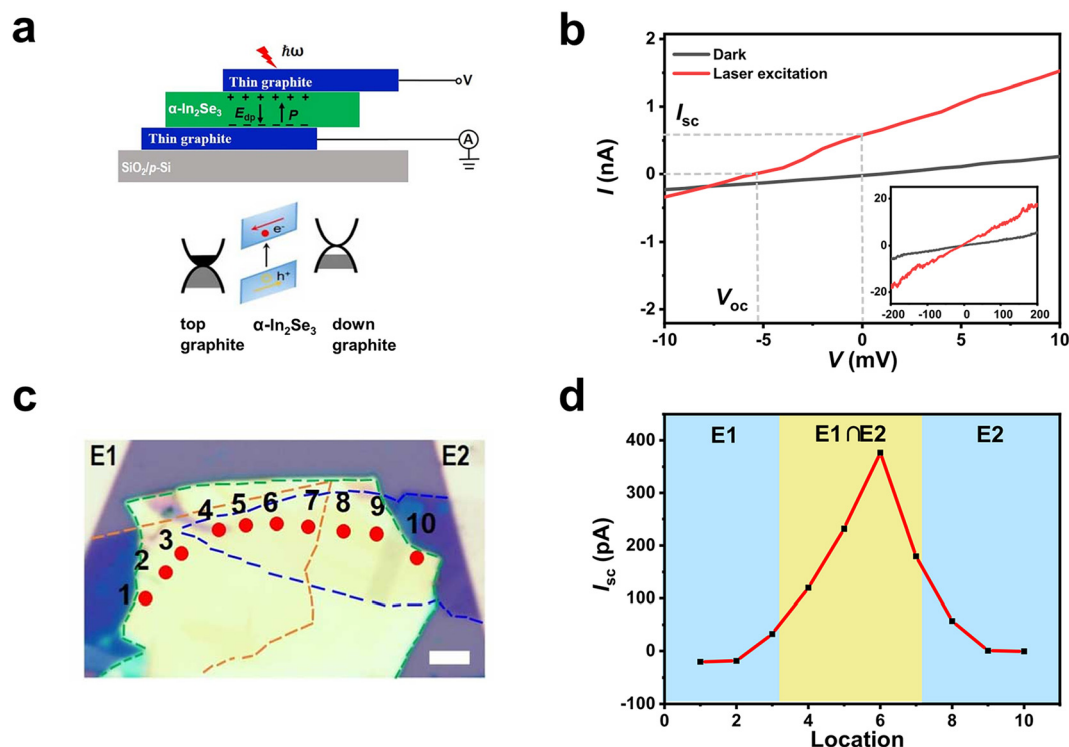


Fig. 2 The BPV effect generation in a 2D α -In₂Se₃ device. (a) Schematic of a 2D α -In₂Se₃ photovoltaic device (D1, up panel) and the corresponding energy band diagram (down panel). (b) I - V characteristics of the D1 device under dark (dark curve) and laser excitation (red curve, 633 nm, 1 μ W). The inset shows an overview of the I - V curve in a wide range. (c) Optical microscopy image of the D1 device, comprising a thin graphite on top (outlined by a yellow dashed line), a 2H α -In₂Se₃ flake (outlined by a green dashed line), and a bottom thin graphite (outlined by a blue dashed line). Scale bar: 5 μ m. Red dots mark laser spots during the measurements. (d) I_{sc} measured at various locations as labeled from #1 to 10 in Fig. 2(c) with a constant laser power of 1 μ W.

0.58 nA is clearly observed with a laser power of 1 μ W, whereas the I_{sc} remains zero under dark conditions. In general, the non-zero I_{sc} is a crucial parameter to evaluate the BPV response. It should be noted that all measurements of I_{sc} and V_{oc} in this work were conducted without any prior poling procedure, ensuring that I_{sc} was merely determined by spontaneous polarization in α -In₂Se₃.

In order to map the distribution of I_{sc} in the 2D α -In₂Se₃ device, we measured the photocurrent at 10 different locations by sequentially moving the laser spot from one graphite electrode (E1) to the overlapping area (E1 ∩ E2) and then to the other graphite electrode (E2), as shown in Fig. 2(c). The photovoltaic response I_{sc} as a function of location (Fig. 2(d)) obviously shows significant non-zero values when the laser spot approaches the overlapping area and reaches its maximum at the center of this region. This can be explained by the finite free path of hot photo-excited carriers in ferroelectric semiconductors. Based on the I_{sc} spatial distribution and the neutralization of the Schottky barrier photovoltaic effect in the overlapping area, we believe that the BPV effect is predominantly responsible for the non-zero I_{sc} in our measurements.

Fig. 3(a) displays a variation in photocurrent I caused by multiple on/off cycles of light illumination, taken from another device D2. Even without bias input, a stable photo-

current I of 4.1 nA is clearly reproduced following the light on/off process with a power of 20 μ W. Here, the photocurrent I exhibits minimal temporal variation over multiple on/off cycles, indicating the reproducibility of the photocurrent output from the BPV device. Fig. 3(b) presents the I - V profiles for laser excitation densities ranging from zero to 1.7×10^3 W cm⁻². The values of I_{sc} and V_{oc} both increase dramatically with the increase of light intensity, with I_{sc} reaching up to 16 nA at a laser intensity of 1.7×10^3 W cm⁻², significantly surpassing that of other materials (will be discussed later in Fig. (4)).

Fig. 3(c) shows V_{oc} and I_{sc} as a function of temperature from 100 to 300 K. V_{oc} of the D2 device increases steadily and reaches saturation at 150 K during the cooling process, while I_{sc} remains relatively constant at different temperatures. The overall resistance, including bulk resistance R_{bulk} and contact resistance $R_{contact}$, exhibits an increase as temperature decreases, according to the I - V curves recorded at various temperatures under dark/light conditions (Fig. S5†). Taking into account such a resistance variation and constant light absorption at different temperatures, a simple equivalent circuit model (Fig. S6†) is proposed to explain the BPV response depending on temperature.⁴¹ In this model, V_{oc} is proportional to R_{bulk} that is determined by the temperature-dependent carrier concentration, while I_{sc} is temperature-inde-

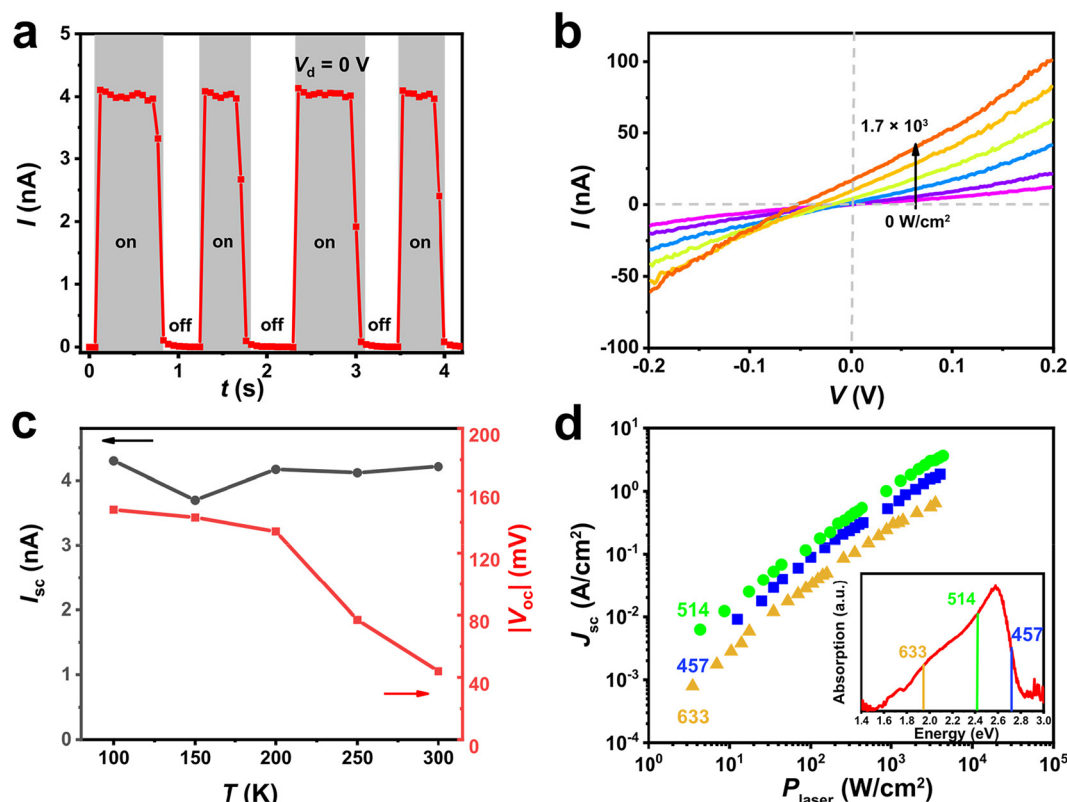


Fig. 3 The BPV response in the other 2D α - In_2Se_3 device (D2). (a) Photocurrent I response recorded by switching light illumination on/off alternatively. (b) I - V curves recorded under various illumination intensities. The inset shows a linear relation between I_{sc} and laser power densities. (c) Temperature dependence of short circuit current I_{sc} (dark circles) and open circuit voltage V_{oc} (red squares). The same laser wavelength of 633 nm and power intensity of 20 μW were applied in measurements for both (a) and (c). (d) Short-circuit current density (J_{sc}) as a function of laser power density (P_{laser}) excited at wavelengths of 457 nm (blue squares), 514 nm (green circles), and 633 nm (red triangles). The inset illustrates the α - In_2Se_3 absorption coefficient as a function of photon energy, with the three excitation wavelengths marked by vertical lines.

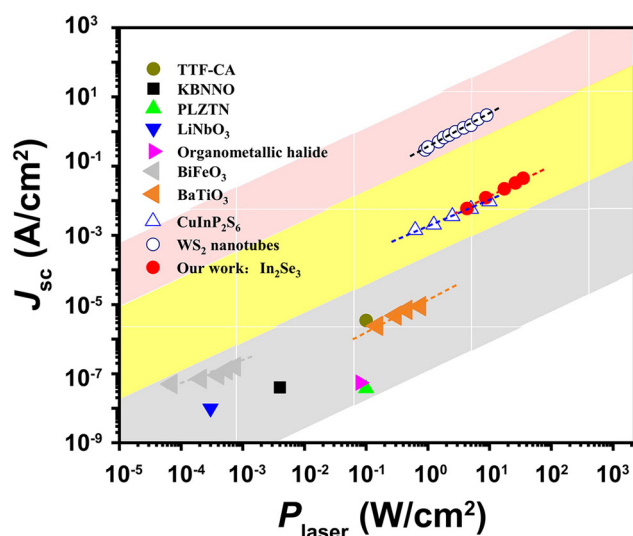


Fig. 4 A summary of the BPV performance of different materials. Data for other materials are taken from the literature (TTF-CA;⁴¹ KBNNO;¹⁴ PLZTN;⁴² LiNbO₃;⁴³ organometallic halide;⁴⁴ BiFeO₃;¹¹ BaTiO₃;¹⁰ CuInP₂S₆;²⁹ and WS₂ nanotubes⁴⁵). Solid red circles represent our experimental results.

pendent for a small R_{contact} (see more details in Section 5, ESI†).

Fig. 3(d) summarizes short-current density J_{sc} as a function of light intensity P_{laser} , calculated from I_{sc} and laser power, and excited at three different wavelengths in the visible region. Here, the value of J_{sc} is linearly proportional to P_{laser} below $\sim 1 \times 10^2 \text{ W cm}^{-2}$, indicating a linear generation of photo-excited electron-hole pairs as light intensity increases. J_{sc} starts to reach saturation at higher power densities (in terms of slope changes), suggesting that the limitations in electron-hole pair separation hinder the increase of short-circuit current. In addition, for the same power density, the 514 nm (2.4 eV) laser generates the highest J_{sc} compared to the 457 nm (2.7 eV) and 633 nm (1.9 eV) lasers. This corresponds to a distinct absorption coefficient for α - In_2Se_3 in the visible light region (as displayed in the inset of Fig. 3(d)). The wavelength of 514 nm (2.4 eV) is close to the resonance peak position in the absorption curve, resulting in a significantly higher BPV photocurrent compared to the other two wavelengths. These results demonstrate that our devices show an outstanding BPV response in the visible light range, whereas conventional ferroelectric materials only exhibit a response in the near-ultraviolet region.

Fig. 4 presents the J_{sc} versus P_{laser} of our devices and other previously reported materials. Here, the data points in the linear region with P_{laser} below $1 \times 10^2 \text{ W cm}^{-2}$ (Fig. 3(d), 514 nm excitation) are taken into account. Obviously, the J_{sc} of $\alpha\text{-In}_2\text{Se}_3$ is nearly two orders of magnitude higher than that of conventional 3D ferroelectric materials (such as perovskite-type bulk oxides, hybrid oxides, organic molecules, and thin films). In addition, $\alpha\text{-In}_2\text{Se}_3$ extends the application of BPV to the visible light region, with J_{sc} comparable to that of 2D ferroelectric CuInP_2S_6 (requires UV light excitation). The J_{sc} of $\alpha\text{-In}_2\text{Se}_3$ is smaller than that of 1D WS_2 nanotubes, due to the improved electrical and optical properties in 1D materials. The polar structure in 1D materials further enhances the BPV capability.¹³ These indicate that the BPV response of the 2D ferroelectric semiconductor $\alpha\text{-In}_2\text{Se}_3$ can bridge the gap between 1D (red shading) and 3D (grey shading) materials.

3. Conclusions

In summary, we report the BPV effect in 2D $\alpha\text{-In}_2\text{Se}_3$ devices. The short-circuit current density of $\alpha\text{-In}_2\text{Se}_3$ is found to be significantly higher than that of conventional ferroelectric 3D materials by nearly two orders of magnitude, owing to the inherent out-of-plane electric polarization in the 2D limit. In addition, $\alpha\text{-In}_2\text{Se}_3$ exhibits an exceptional BPV response in the visible light region, owing to its narrow bandgap structure. This research highlights the potential of utilizing 2D ferroelectric semiconductors for advanced optoelectronic applications such as solar cells and photodetectors.

4. Materials and methods

4.1 Device fabrication

Single crystals of $\alpha\text{-In}_2\text{Se}_3$ and bulk graphite used in this work were provided by HQ Graphene and NGS Naturgraphit Companies, respectively. The $\alpha\text{-In}_2\text{Se}_3$ flakes and thin graphite were successively exfoliated and transferred to a $\text{SiO}_2/\text{p-Si}$ substrate. Then, thin graphite/ $\alpha\text{-In}_2\text{Se}_3$ /thin graphite heterostructures were obtained through an all-dry transfer technique by using a polydimethylsiloxane (PDMS) stamp in a N_2 atmosphere glove box. Metal electrodes were obtained by using a laser direct write patterning technique, followed by electron beam evaporation of Ti/Au (5 nm/30 nm). The samples were subsequently immersed in NMP at 80 °C and maintained for 1 h during the lift-off process. Finally, these electrodes were processed with ozone at 80 °C for 10 min.

4.2 AFM, Raman spectra, and SHG characterization

The surface morphology of the device was characterized using an atomic force microscope (AFM, Dimension Icon-GB, Bruker). Raman spectra were recorded using a micro-Raman system (Horiba T64000) equipped with a 514 nm excitation laser. To prevent sample degradation, the laser power was set lower than 0.1 mW. In addition, the system was calibrated

using the Raman peak of Si at 520 cm^{-1} prior to measurements. For SHG spectroscopy, the excitation light came from a mode-locked Ti:sapphire laser with a central wavelength of $\sim 800 \text{ nm}$, 76 MHz repetition rate, 150 fs pulse duration, and an average power of 40 μW . Using a 50 $\times$ objective lens and a backscattering configuration, the laser was focused on the sample surface. The SHG signal was collected using an air-cooled CCD spectrometer. For the measurements of in-plane polarization, the excitation laser was focused at normal incidence on the sample surface. To measure the contribution of out-of-plane polarization, the excitation laser was focused at a tilted angle (approximately $15^\circ\text{--}30^\circ$). The transmissivity of the optical system was carefully calibrated to determine the absolute power level at the focal plane. The emission spectra were normalized based on the square of the focal power.

4.3 Electrical transport and photovoltaic measurements

Electrical transport measurements were conducted at room temperature under vacuum. The I - V characteristics under the dark conditions were measured using a Keysight source meter (Keithley 4200 Semiconductor parameter analyzer) connected to a cryogenic probe station (CRX-VF, Lake-Shore).

Photoelectric measurements were conducted using an optical cryostat (Microscopy Cryostat System N_2 , Cryo Industries) under high-vacuum conditions. These devices were illuminated using three different lasers with wavelengths of 633 nm (He-Ne laser), 514 nm, and 457 nm (two diode lasers). The laser spot sizes were approximately 2.5 μm (633 nm), 1.7 μm (514 nm) and 1.6 μm (457 nm). The laser was focused on samples using a 50 $\times$ objective lens. The applied bias voltage and the output current were controlled *via* a Keithley 2614B electrometer that was connected to the optical cryostat *via* feedthroughs. The open-circuit voltage and short-circuit current were derived from the current-voltage curves. The photocurrent response (I - t curve) was obtained using a manually controlled chopper with a blade switching the laser beam between blocked and open states. The temperature-dependent I - V measurements were carried out using an optical cryostat with a heating source.

Data availability

The data that support the findings of this work are available from the corresponding author upon reasonable request.

Conflicts of interest

The authors declare no conflicts of interest.

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